Log2: 25th May 2026 (2.36pm)

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1. Objective of This Session

The goal of this session was to re-check department GPU server availability after the previous attempt.

Specifically, the session tested:

- SSH access to Ampere
- SSH access to Ada
- GUI/X11 forwarding
- ROS 2 availability
- Gazebo availability
- GPU availability
- Singularity/Apptainer availability
- Storage layout
- OpenGL rendering behavior

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2. Updated Server Status

Server	Previous Status	Today's Status	Assessment
Ada	Not working	Working	Now usable
Ampere	Working	Not working	DNS/hostname resolution issue
Turing	Not primary	Working as jump host	Useful intermediate server
Tesla	Working login	Cannot forward to Ampere	Still usable as entry point

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3. Ampere Connectivity Failure

Commands Used

```
ssh -X e21283@ampere.ce.pdn.ac.lk
```

Output

```
ssh: Could not resolve hostname ampere.ce.pdn.ac.lk: Name o
r service not known
```

Also observed earlier:

```
channel 0: open failed: connect failed: Name or service not
known
stdio forwarding failed
kex_exchange_identification: Connection closed by remote ho
st
Connection closed by UNKNOWN port 65535
```

Conclusion

Ampere is currently not reachable by hostname.

This is likely not a password or account issue. The error indicates a DNS/name-resolution problem or server/network-side issue.

Possible causes:

- Ampere hostname temporarily removed from DNS
- Ampere is down
- Internal routing issue
- Tesla cannot currently resolve Ampere
- Server-side SSH or network configuration problem

Ampere should be revisited later.

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4. Successful Access to Turing

Command Used

```
ssh -X e21283@turing.ce.pdn.ac.lk
```

Result

Login succeeded.

Turing is available and can be used as an intermediate access point.

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5. Successful Access to Ada

Command Used from Turing

```
ssh -X e21283@ada.ce.pdn.ac.lk
```

Result

Login succeeded.

Ada is now working, even though it was not working during the previous session.

Conclusion

Ada has now become the primary usable server for today’s testing.

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6. X11 / GUI Forwarding Verification on Ada

Command Used

```
xclock
```

Output

```
Warning: Missing charsets in String to FontSet conversion
```

Conclusion

The warning is not critical. Since `xclock` launched, X11 forwarding is working.

This confirms:

- MobaXterm X server works
- SSH X11 forwarding works through Turing → Ada
- Remote GUI applications can be displayed locally

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7. ROS 2 Availability Check on Ada

Command Used

```
ros2 --version
```

Output

```
ros2: command not found
```

Conclusion

ROS 2 is not installed system-wide on Ada.

This is not a blocker because containerization is available through Apptainer/Singularity.

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8. Gazebo Availability Check on Ada

Command Used

```
gazebo --version
```

Output

```
gazebo: command not found
```

Conclusion

Gazebo is not installed system-wide on Ada.

Same as Ampere: Gazebo should be installed inside a container, not directly on the server.

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9. GPU Verification on Ada

Command Used

```
nvidia-smi
```

Result

Detected:

- 3 × NVIDIA RTX 6000 Ada Generation GPUs
- Driver Version: 580.82.07
- CUDA Version: 13.0
- GPU memory: 49140 MiB per GPU

Conclusion

Ada has very strong GPU resources and is suitable for future GPU workloads.

However, GPUs are already being used by other users, so this is a shared research server.

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10. Singularity / Apptainer Verification

Command Used

```
singularity --version
```

Output

```
apptainer version 1.4.5
```

Conclusion

Apptainer is installed and working.

This is important because the robotics stack can be deployed through a container.

Recommended approach:

```
Ada / Turing
↓
Apptainer container
↓
Ubuntu 22.04
↓
ROS 2 Humble
↓
Gazebo + RViz2
```

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11. Storage Inspection on Ada

Command Used

```
df -h
```

Key Findings

Available storage:

```
/dev/sda1    9.1T    2.2T    6.4T    26%    /scratch1
```

Also observed:

```
10.40.18.1:/export/home    11T    11T    78G    100%    /home
```

Conclusion

The `/home` partition is full.

Project work should not be done inside `/home`.

Recommended workspace:

```
/scratch1/e21283/go2_project
```

Create it using:

```
mkdir -p /scratch1/$USER/go2_project
cd /scratch1/$USER/go2_project
```

12. `/storage` Check

Command Used

```
ls /storage
```

Output

```
ls: cannot access '/storage': No such file or directory
```

Conclusion

Unlike Ampere, Ada does not expose scratch storage under `/storage`.

On Ada, the correct scratch path appears to be:

```
/scratch1
```

13. OpenGL Verification

Command Used

```
glxinfo | grep OpenGL
```

Output Summary

```
OpenGL vendor string: Mesa
OpenGL renderer string: llvmpipe
OpenGL version string: 4.5
```

Conclusion

OpenGL rendering is using Mesa llvmpipe, meaning CPU software rendering.

This is expected over normal SSH X11 forwarding.

Important distinction:

```
GPU compute: Working
Remote OpenGL GPU rendering: Not working
```

This is not a failure for now.

It only becomes important when running heavy Gazebo/RViz scenes.

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14. Overall Conclusion

Today's session changed the server plan.

Previously:

```
Ampere = primary working server
Ada = not accessible
```

Now:

```
Ada = working
Ampere = currently not resolving
```

Ada is suitable for continuing the project because it has:

- SSH access
- X11 forwarding
- RTX 6000 Ada GPUs
- CUDA-visible GPUs
- Apptainer container support
- Large `/scratch1` storage

Missing components:

- ROS 2
- Gazebo

These should be handled inside a container.

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15. Updated Immediate Next Steps

Step 1 — Create project workspace on Ada

```
mkdir -p /scratch1/$USER/go2_project
cd /scratch1/$USER/go2_project
```

Step 2 — Use Apptainer instead of system ROS

Target container:

```
Ubuntu 22.04
ROS 2 Humble
Gazebo
RViz2
TurtleBot3 packages
```

Step 3 — Validate basic ROS 2 simulation

First validation target:

```
TurtleBot3 Gazebo demo
```

Required checks:

```
ros2 topic list
ros2 topic hz /scan
rviz2
```

Step 4 — Recheck Ampere later

Run later from Tesla or Turing:

```
ssh -X e21283@ampere.ce.pdn.ac.lk
```

If it still fails, report:

```
ampere.ce.pdn.ac.lk cannot be resolved from Tesla/Turing.
DNS or server-side access may need checking.
```

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16. Strategic Outcome

The project is still technically feasible.

The validated server has changed from Ampere to Ada for now.

Current recommended workflow:

```
Windows 11
↓
MobaXterm SSH/X11
↓
Tesla or Turing
↓
Ada
↓
/scratch1/$USER/go2_project
↓
Apptainer container
↓
ROS 2 Humble + Gazebo
↓
Go2 simulation
```